WMT Practical Lab Test – 1 Hour

Duration: 1 Hour

Total Marks: 100

📌 Instructions:

- Complete all tasks
- You may use any framework (Node.js / PHP / Python)
- Internet allowed for documentation
- Final output must be **working + hosted OR runnable locally**

Task: Build & Deploy a Simple Login System

🔹 Part 1 – Frontend (25 Marks)

Create a **Login Web Page**

Requirements:

- Use **HTML + CSS**
- Fields:
  - Email
  - Password
- Button:
  - Login

Validation:

- If empty → show message:

  "Please fill all fields"

🔹 Part 2 – Backend API (30 Marks)

Create a backend server with an API:

Endpoint:

```
POST /login
```

Requirements:

- Receive email & password
- Validate input
- Send response:
  - ✅ "Login Successful"
  - ❌ "Invalid Credentials"

🔹 Part 3 – Database (20 Marks)

- Create a database (any):
  - MySQL / MongoDB / Firebase

Table/Collection:

```
Users
- id
- email
- password
```

Requirement:

- Insert at least **1 user manually**
- Backend must check login using this data

🔹 Part 4 – Integration (15 Marks)

- Connect frontend → backend API
- Use:
  - Fetch API / Axios

Requirement:

- On button click:
  - Send request to API
  - Display response message on page

🔹 Part 5 – Deployment (10 Marks)

- Deploy your app using:
  - Vercel / Netlify / Firebase / Railway / AWS

👉 OR (if no time):

- Run locally and show working system

Bonus (+10 Marks)

- After successful login:
  - Redirect to **dashboard.html**
  - Show:

```
Welcome, [email]
```

Web & Mobile Technologies (WMT) – Practical Examination

⏱️ Duration: 1 Hour

🎯 Total Marks: 100

General Instructions

- Answer **ALL questions**
- Use any suitable technology stack:
  - (Node.js / PHP / Python / Firebase, etc.)
- Internet usage is allowed for documentation only
- Your final solution must be:
  - ✅ Functionally working
  - ✅ Well-structured
- Viva may be conducted based on your implementation

Question 1 – Web Application Development (100 Marks)

Design and implement a **User Authentication System** with the following specifications:

🔹 1. Frontend Interface (20 Marks)

Develop a **Login Page** with:

- Input Fields:
  - Email
  - Password
- A Login button

Requirements:

- Use HTML, CSS, and optional JavaScript
- Apply basic styling for usability

- Display appropriate error messages when:
  - Fields are empty
  - Invalid input is entered

🔹 2. Backend Development (25 Marks)

Develop a backend service that:

- Provides an API endpoint:

```
POST /login
```

-
- Accepts user credentials (email, password)
- Validates incoming data
- Returns appropriate responses:
  - Success (valid credentials)
  - Failure (invalid credentials)

🔹 3. Database Implementation (20 Marks)

Design and connect to a database:

Structure:

```
Users Table / Collection:
- id
- email
- password
```

Requirements:

- Insert at least one user record

- Backend must verify login using stored data

🔹 4. System Integration (15 Marks)

- Connect frontend with backend API
- Use:
  - Fetch API / Axios / equivalent
- Display server responses dynamically on UI

🔹 5. Deployment / Execution (10 Marks)

- Deploy application using:
  - AWS / Firebase / Vercel / Netlify / Railway

OR

- Run locally and demonstrate full functionality

🔹 6. Code Quality & Structure (10 Marks)

Marks will be awarded for:

- Clean and readable code
- Proper file structure
- Logical implementation

Bonus Task (+10 Marks)

- After successful login:
  - Redirect user to a new page (Dashboard)
  - Display a personalized message:

```
Welcome, [User Email]
```